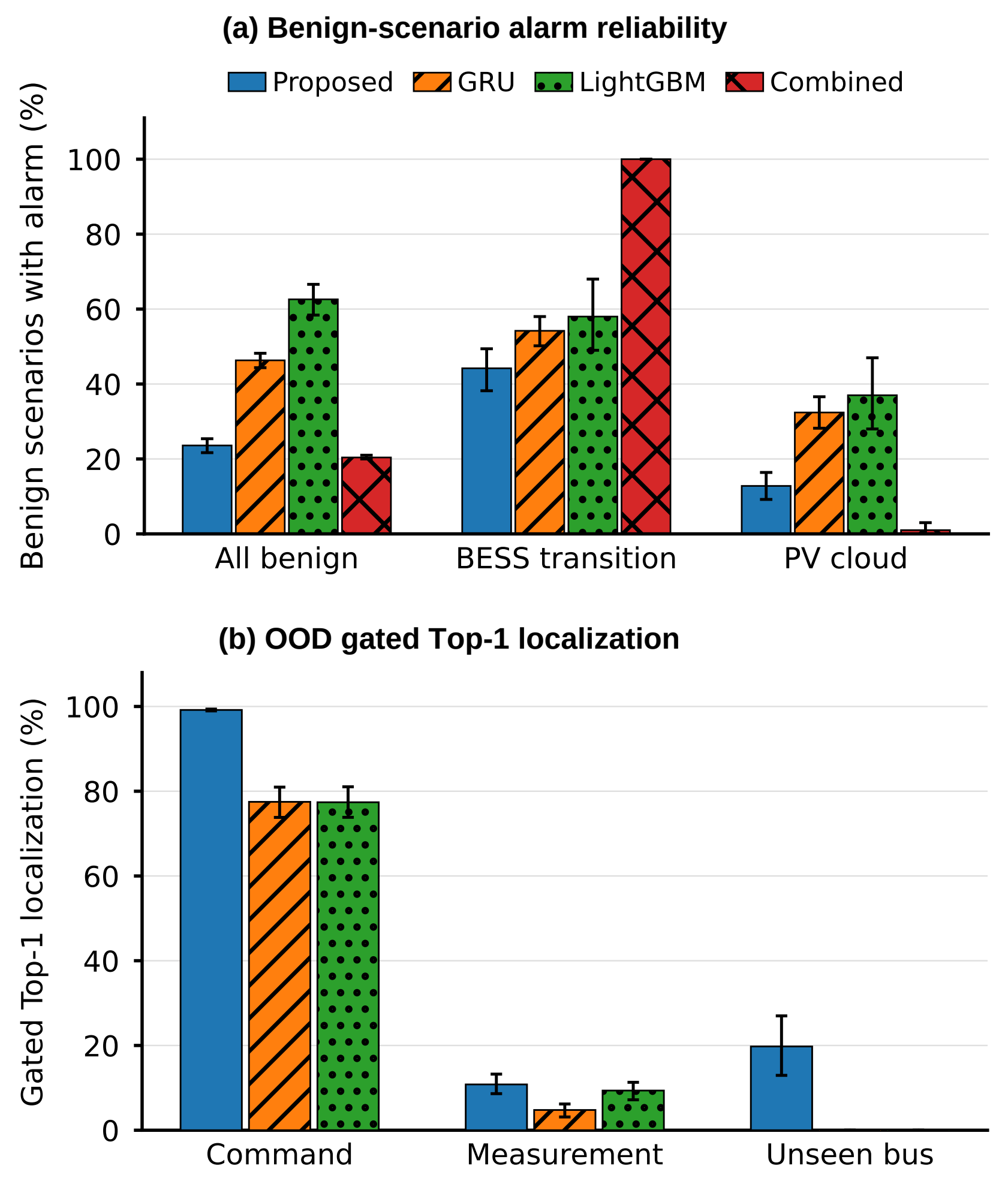

(a) Benign-scenario alarm reliability
Proposed
GRU
LightGBM
Combined
Benign scenarios with alarm (%)
100
80
60
40
20
0
All benign
BESS transition
PV cloud
(b) OOD gated Top-1 localization
Gated Top-1 localization (%)
100
80
60
40
20
0
Command
Measurement
Unseen bus